

Building a predictive model for analyzing customer churn in the Telecom industry.

Business Problem and Objective

Customer churn is a major issue in the Telecom industry. It can be described as customers who stop using the company's services within a defined period. Because telecom services are typically subscription-based, churn is a critical performance metric tied directly to revenue and growth. The objective of this report is to build a predictive model (logistic regression) for labeled churn data. We have also applied survival analysis to the data. The data has been accessed from Kaggle. The analysis has been done using SAS.

Data Explanation

The dataset initially contains 33 variables from CostumerID to Churn value (binary) to churn score and then churn reason. The churn value is the dependent variable which will be used in logistic regression. Churn reason is customer's specific reason for leaving the company. Directly related to Churn Category, whereas churn score is a value from 0-100 that is calculated using the predictive tool IBM SPSS Modeler. The model incorporates multiple factors known to cause churn. The higher the score, the more likely the customer will churn.

Data Preprocessing and Feature Engineering

First, we checked for duplicate values and found that there are no duplicate values. Then, we checked for extreme values using proc means procedure. After that, we produced several new variables using feature engineering. Some of the newly created variables are: TotalCharges_num, Paperless_Flag, Partner_Flag, Dependents_Flag, churn_flag, Internet_Type, Payment_Group, Tenure_Group, AvgChargePerMonth, High_Monthly_Charge, tenure_m, monthlycharges, contract, and paymentmethod.

Logistic Regression

Logistic regression is applied using the target variable as 'Churn value'. Its value is either 1 - the customer left the company this quarter, or, 0 - the customer remained with the company.

The logistic regression indicates that customer churn is strongly influenced by tenure and monthly charges. Customers who have been with the company longer are significantly less

likely to churn, while higher monthly charges increase churn risk. Total lifetime spending does not add strong predictive value after accounting for tenure. These results suggest that early-stage customers and high-bill customers should be prioritized for retention strategies.

Survival Analysis

Survival analysis was applied to understand both the timing and drivers of customer churn. Results indicate that churn risk is highest during the early months of the customer's lifecycle and increases with higher monthly charges. As customer tenure increases, churn risk declines significantly, highlighting the importance of early engagement. These findings suggest that retention strategies should focus on onboarding, pricing sensitivity, and timely intervention. By leveraging survival analysis, the business can proactively identify high-risk customers and implement targeted actions to reduce churn and improve long-term customer value.

Key Findings

Logistic regression was used to identify **which factors increase or decrease the likelihood of customer churn**, without explicitly modeling time.

Key Findings

Customer Tenure Is the Strongest Retention Indicator

- Customers with longer tenure are **significantly less likely to churn**.
- This indicates that churn risk is highest during the **early stages** of the customer relationship.

Higher Monthly Charges Increase Churn Risk

- Customers with higher monthly bills show a **greater probability of churn**, suggesting strong price sensitivity.
- Billing pressure is a major contributor to customer dissatisfaction and attrition.

Lifetime Spend Adds Limited Additional Insight

- Total charges provide limited incremental value once tenure is accounted for.
- This suggests that **how long** a customer stays matters more than **how much** they have spent.

Overall Insight:

Churn is strongly influenced by tenure and pricing, with early-stage and high-cost customers being most at risk.

◆ Key Findings — Survival Analysis (Time-to-Churn)

Purpose Recap

Survival analysis was used to understand **when customers churn** and how churn risk evolves over time, while accounting for active (censored) customers.

Key Findings

Churn Risk Is Highest Early in the Customer Lifecycle

- Survival curves show a **steep decline in retention during the initial months**.
- Customers who survive beyond this early period are substantially more likely to remain long-term.

Pricing Influences Not Just Churn, but Churn Timing

- Higher monthly charges accelerate churn, causing customers to leave **earlier** rather than later.
- Pricing pressure affects both **probability and speed** of churn.

Retention Improves Significantly With Time

- As tenure increases, churn risk declines, indicating growing customer loyalty and switching costs over time.
- Long-tenured customers represent a **more stable and valuable segment**.

Overall Insight:

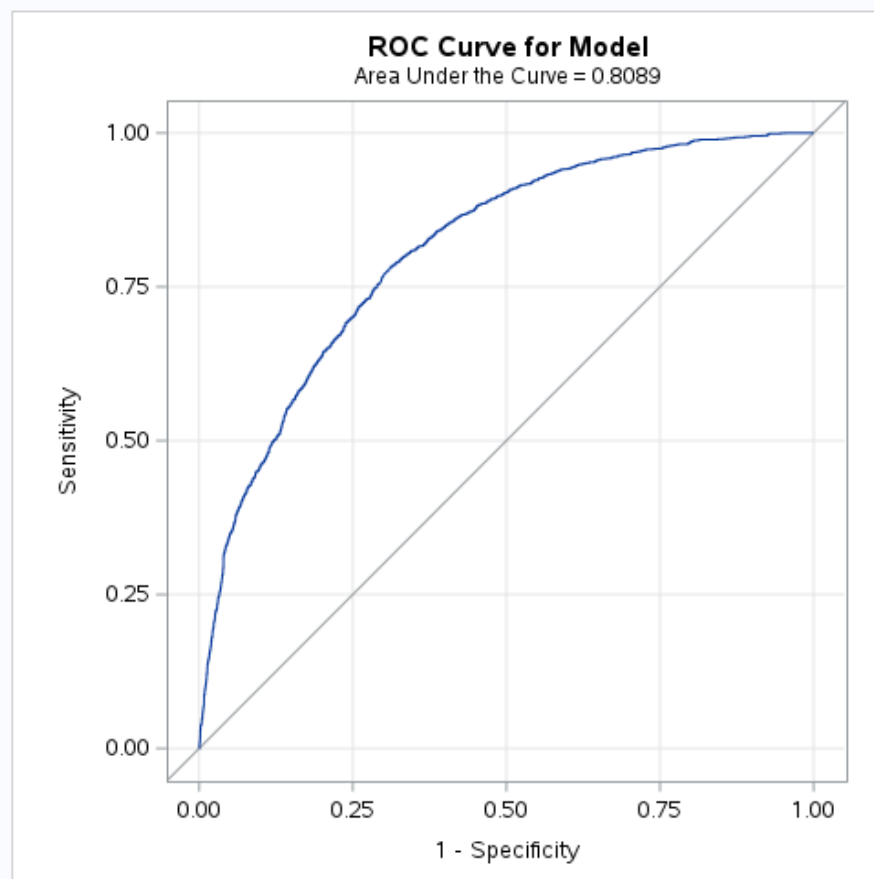
Churn is a time-dependent process, with early tenure and high monthly charges accelerating customer attrition.

The FREQ Procedure

Frequency

Table of Churn Value by pred_churn			
Churn Value(Churn Value)	pred_churn		Total
	0	1	
0	4704	470	5174
1	1042	827	1869
Total	5746	1297	7043

Pairs	9049047	c	0.809
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Business Impact

The analysis enables the business to proactively manage customer churn by identifying both the key drivers and the timing of customer attrition. By focusing on early-tenure and high-cost customers, optimizing the timing of retention interventions, and applying targeted pricing strategies, the organization can improve customer lifetime value, reduce revenue loss, and increase the overall effectiveness of retention initiatives.